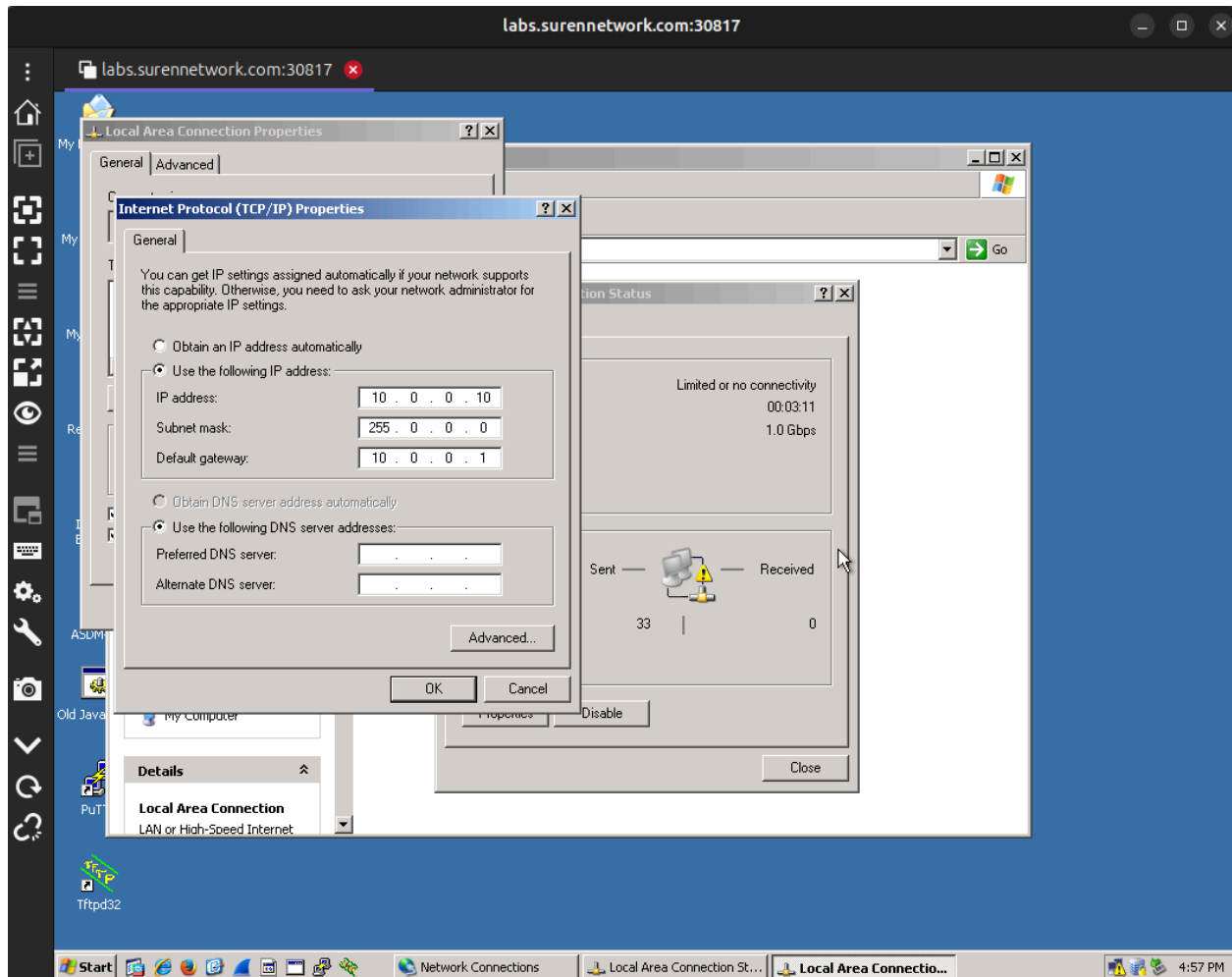
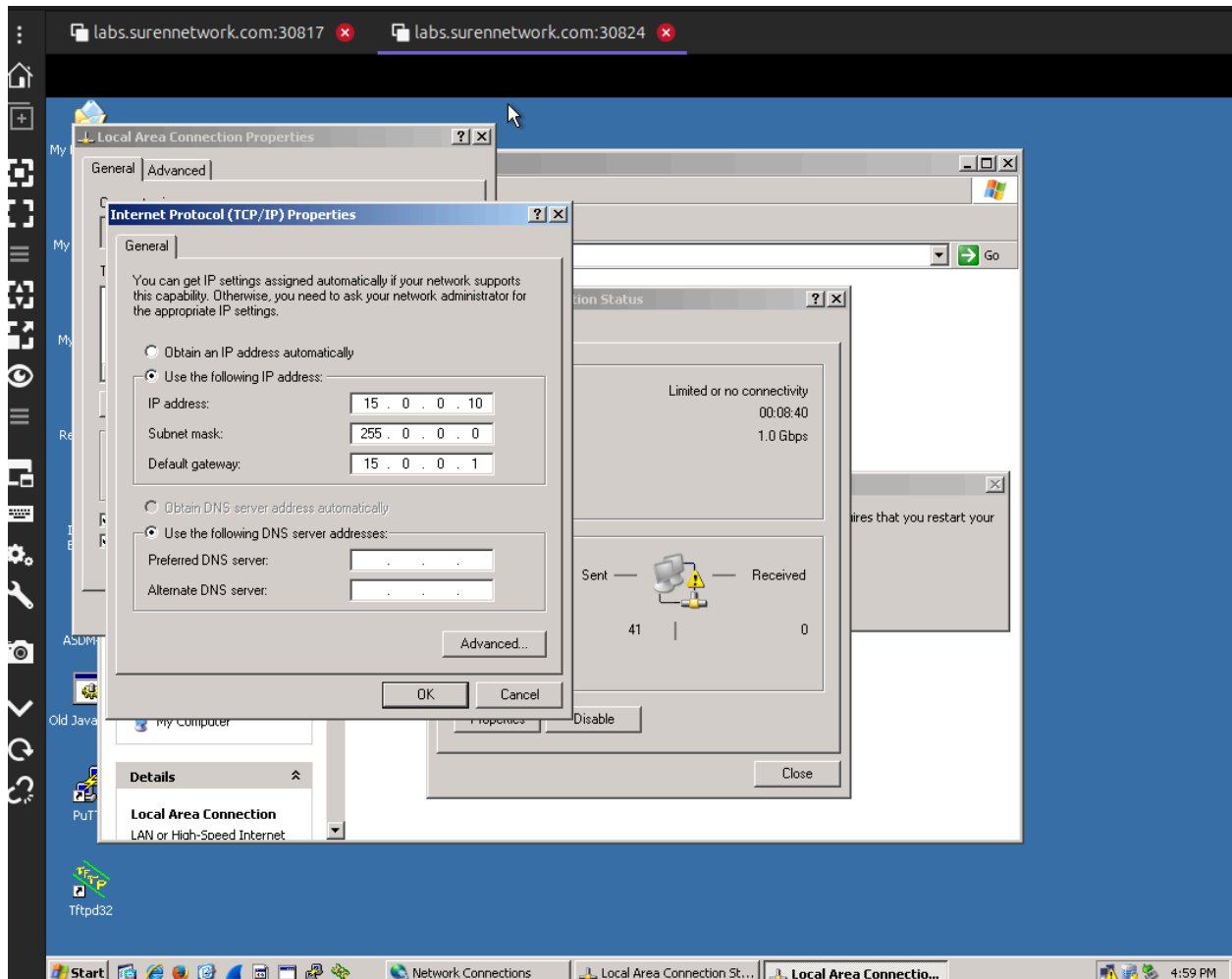
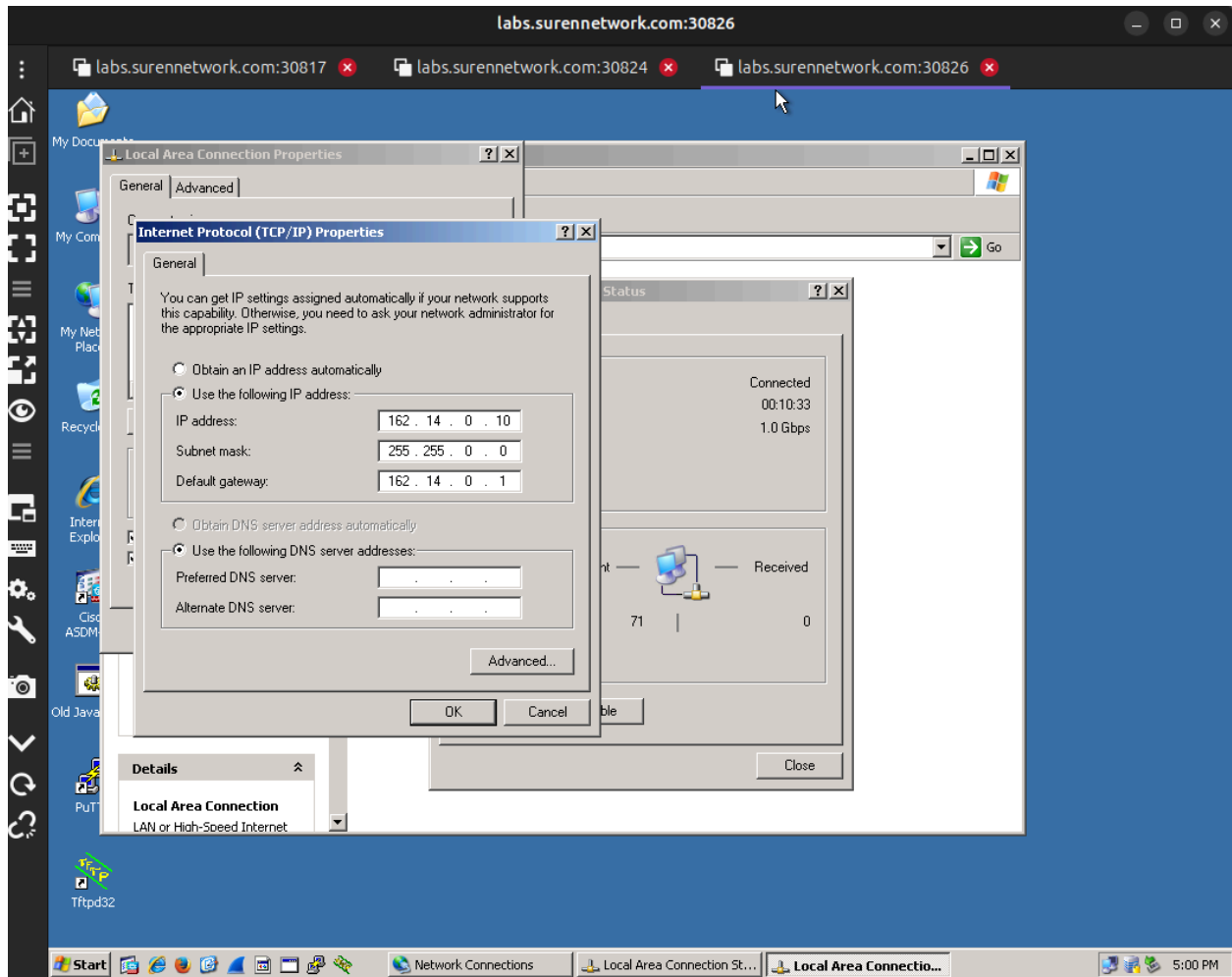








Press RETURN to get started.

R1#

R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
S       172.19.0.0/16 [1/0] via 20.0.0.2
S       192.168.168.0/24 [1/0] via 20.0.0.2
```

R1#

```
R2
R2(config)#ip route 21.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 172.19.0.0 255.255.0.0 30.0.0.2
R2(config)#ip route 15.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 192.168.168.0 255.255.255.0 30.0.0.2
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
R2(config)#exit
R2#
*Aug 22 16:15:28.250: %SYS-5-CONFIG_I: Configured from console by console
R2#
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/0
L        20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/1
L        30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
S    172.19.0.0/16 [1/0] via 30.0.0.2
S    192.168.168.0/24 [1/0] via 30.0.0.2
R2#
```

```
R3(config)#ip route 192.168.168.0 255.255.255.0 21.0.0.2
R3(config)#ip route 162.14.0.0 255.255.0.0 21.0.0.2
R3(config)#exit
R3#
R3#
*Aug 22 16:20:37.314: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/2
L    21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/0
L    30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
R3#
```

```
R4
R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1
R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1
R4(config)#exit
R4#
R4#
*Aug 22 16:25:38.020: %SYS-5-CONFIG_I: Configured from console by console
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
S    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
S    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
R4#
```

```
R5
R5(config)#ip route 10.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 20.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 30.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 172.19.0.0 255.255.0.0 21.0.0.1
R5(config)#ip route 15.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2
R5(config)#exit
R5#e
*Aug 22 16:30:48.616: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/0
L     21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
     192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
L     192.168.168.1/32 is directly connected, Ethernet0/1
R5#
```

```
R6(config)#ip route 30.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 172.19.0.0 255.255.0.0 192.168.168.1
R6(config)#ip route 21.0.0.0 255.0.0.0 192.168.168.1
R6(config)#ip route 15.0.0.0 255.0.0.0 192.168.168.1
R6(config)#exit
R6#
R6#
*Aug 22 16:38:05.659: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    162.14.0.0/16 is directly connected, Ethernet0/0
L    162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.2/32 is directly connected, Ethernet0/1
R6#
```

